

3.3 Circular motion

Further Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

For circular movement, acceleration points inward with magnitude $v^2/r = \omega^2 r$. In vertical circles, radial resultant changes with position because the radial component of weight changes. State the chosen positive radial direction before writing Newton's second law.

Worked example: follow the reasoning

At bottom of vertical circle, $T - mg = mv^2/r$; at top, $T + mg = mv^2/r$ for a taut string.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 0.5 kg mass travels at 4 m s^{-1} around radius 2 m on a horizontal frictionless table. [3]
Determine string tension.

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Radial acceleration $v^2/r = 16/2 = 8 \text{ m s}^{-2}$ [1]; tension supplies resultant [1]; $T = ma = 4 \text{ N}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.